

MAY 1941

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STORIES OF SUPER TIME AND SPACE

Locked in Ice
Caverns on Neptune
in
ICE PLANET
by Carl Selwyn

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DERELICTS OF URANUS
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**WHEN TIME ROLLED
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*** START OF THE PROJECT GUTENBERG EBOOK THE FACTS OF
LIFE ***

The FACTS of LIFE

by P. SCHUYLER MILLER

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"The ability to profit by past experience and to use this knowledge as a guide to future action may, ladies and gentlemen, be taken as the primary differentiation between the animal and the vegetable kingdoms."

Thus Professor Melchizedek Hobbs, principal of the Springville Free Academy, on the day long-gone when I began my higher education. I can see him yet, the apotheosis of the Victorian schoolmaster, Ichabod Crane, come to life: the sparse, sandy hair brushed carefully across his bony skull, his long nose trembling with the vehemence of his argument, his artist's fingers stained with the chemicals which he had lately been preparing in the school's laboratory, fumbling nervously with his mauve cravat and peering worriedly over the tops of his steel-bowed spectacles at our bright and shining faces.

To Mr. Melchizedek Hobbs every moment of every day was precious. Those of us who came to know him a little more intimately in the four years that followed realized that he was not like other teachers. His teaching was the driving purpose of his life, second only to the keen and insatiable curiosity which sent his vulturine nose prying into the intimacies of Nature and ferreting out improbable facts to the greater glory of botanical science. Now, on our first day at the Academy, he paced the rostrum like a moulting crane, wholly intent on the seriousness of his peroration.

Honeyed persuasion was in his voice, and a note of steel when it was needed, for by any standards Mr. Melchizedek Hobbs was no mean orator. Now he made an appeal to our young emotions:

"How often in one's journeyings is the heart warmed and the spirit moved by the solicitude shown by even the lowliest of God's thinking creatures in the care and upbringing of its young! How appalling is the contrasting lethargy which characterizes the race of the cabbage and the vegetable marrow! With what wanton abandon does the profligate thistle scatter its plumed seeds to the four winds, yet with what loving patience does the gentle hind nurture her fawn and bring it to maturity.

"Education, ladies and gentlemen, is not the prerogative of Mankind! The kitten learns from the wise mouser, its mother, to stalk its wary prey. The sparrow in its nest is taught to spread its trembling wings. Even the field

mouse learns to know its natural enemies and to recognize them from afar. It is God's will on Earth that in every thinking race the parent should instruct its young, the adult impart the accumulated wisdom of its kind to the immature. Education, ladies and gentlemen, is the heritage of the animal kingdom—the privilege which divides us from the leek and the asparagus! I trust that you will not deny that heritage!"

Thus Mr. Melchizedek Hobbs, in the days when I first knew him. There were a few of us who tagged him through the woods and fields, listening to his painfully erudite disquisitions on matters of botany or zoology, following his kicking heels and flying coat-tails in wholly undignified pursuit of some new butterfly or beetle, or laboring home under the weight of collecting boxes stuffed with mosses and rare ferns. We learned little enough, I suppose, for I find it hard now to distinguish a primrose from a cowslip, but we appreciated the very real enthusiasm which was his, and his sincere desire to learn and to impart what he had learned.

Then, in our turn, we graduated and went our separate ways. I heard that a maiden aunt in England—some forgotten relative of his mother's—had died and left Professor Hobbs an income which permitted him to leave the Academy and open a little greenhouse which was as much a laboratory as a business enterprise. I wrote him a letter of congratulation, and from time to time in my wanderings I sent him slips of rare or beautiful plants which came to my attention. And then, only a few months before my travels were ended and I came back to Springville, I happened on the Zulu rose.

Where it got its name I do not know, for to the best of my knowledge there are not and never have been Zulus in Madagascar. Probably some African explorer, a little off his regular course, paid a fleeting visit to the isle of marvels and bestowed his taxonomic benediction on everything that came to his attention. In any case, and by any name, the Zulu rose would be the same anomaly.

I had gone to Madagascar with some wild idea of finding and dragging back to civilization the fabled man-eating tree. That I failed was probably due in part to the fact that it never existed, save in some retired colonel's fevered

imagination. I panted off on the trail of the Aepyornis and had to be satisfied with a much addled egg, still on display in the Springville Free Museum and Loan Library. I shot lemurs and hunted for missing links, for Darwin's "Origin of Species" had been very much before the undergraduate eye during my college career. All I found, in the end, was the Zulu rose.

What first attracted me to the plant was the fact that it was never twice the same. There was a family likeness—about as much as there is between me and my brother Charles—but that was as far as it went. No self-respecting plant behaves like that.

The first that I saw was in a young lady's hair, and I only noticed in passing that it was very much like a full-blown rose, with crimson, satiny petals. The following morning, on my way back to the hotel, I saw the same rather spectacular blossom in a private garden and was somewhat puzzled by the fact that it was growing on a stalk very much like an Easter lily, with long, swordlike leaves in a whorl about its base. There were several colors on the same bed—reds and creamy whites and one lot of a striking orange color.

Then, in the forest, I found the things growing in an entirely different manner. At least, the crotchety old duffer who was guiding me swore that they were the same plant, although these were growing like parasitic orchids on huge mats of threadlike roots. The petals were more orchid-like, too, and less flamboyantly colored, and I assumed that this might be an ancestral form from which the cultivated varieties had been developed.

All in all, I think I saw some twenty different varieties of Zulu rose and no two of them were alike. That I did not see the one thing that was of importance, or even hear of it, can be ascribed only to the notoriously bad luck of the Abercrombies. I saw Zulu roses that were like thistles, and others that were like sunflowers. I saw them growing like water-lilies, like cactus, and like edelweiss. They weren't common, but wherever they were they seemed to be perfectly adapted to the environment they were in. Their perfume was really overpowering and not entirely pleasant, and I noted in passing that there were never any bees or other insects near them. Unfortunately, while I mentioned the fact to my old teacher in the letter I sent with cuttings of three or four of the plant's many varieties, I let it go at that.

Nearly a year passed before I saw Miss Liberty's torch raised over New York harbor and watched the friendly hills of the Mohawk Valley closing in on either side of the train. Springville was just what it had been fifteen years before—the same rutted streets, the same fly-specked store windows, the same sleepy horses in front of the Oriskany House—even the same sparrows quarreling under the eaves of the Methodist Church. Jim Selford hacked me up from the station—he's Mayor of Springville now, and proprietor of the garage which he opened with much misgiving when he was sure that the horse had gone to stay. In the course of our parade up Main Street he gave me thumbnail sketches of practically everyone of importance who had been born, died, or come to fame since I left town.

I had my first hint that all was not well when we passed the hole-in-the-wall that had, during my childhood, been a combined tobacco and sweet shop. It had an already weather-beaten sign over the door—"HOBBS—FLORIST"—and busy about the front of the shop was a familiar figure in the normal costume of a respectable upstate female.

Jim cast a glance over his shoulder at my question. "Her? That's Abigail Jones; tends for old Hobbs." He spat accurately at the iron hitching post in front of the First National Bank.

Now I know Jim Selford. The boys I cronied with had spent a good deal of their time around his livery stable, and our own yard had backed up on his. There had been certain disagreements about the uses to which his pears should be put, if I remember. At any rate, I knew he was holding something back.

"How is Professor Hobbs?" I inquired innocently. "I suppose he's one of the city fathers by now."

Jim looked at me with suspicion, but I kept an impassive face. He uncrossed his legs, crossed them again, picked up the whip and gave the bay mare a cut across the rump that made her jump. "Geeup!" he answered.

I recognized the gambit. I must give before I would get. "Has he had any luck with the plants I sent him from abroad?" I asked. "There were some

very rare ones that you won't find in any of the big botanical gardens. If he can grow them here, it ought to put Springville on the map."

That did it. Jim planted both feet with a clump and twisted the reins around the whip. He spat his quid into the gutter, dusted off the plug, and cut a new chaw. Then he turned on me.

"You're into it too, are you? Might of knowed! If there was ever a worse show an' hullabaloo than that old fool has raised I never seen it. If I was the Widder Jones I'd starve afore I'd leave my daughter tend shop for the kind he is. Batty—that's what's wrong with him! Crazy as a coot! And dangerous! Them damn flowers! Ptah!"

Then he closed up like a clam. I got not one word more out of him until we pulled up in front of my uncle's house, now mine. Then: "Go on up there," he said. "See for yourself. Giddap!"

Which, of course, is exactly what I did. Of all my old friends and cronies, Melchizedek Hobbs was the one to whom I had been closest. Jeremiah Jones had written me a few times from Chicago, where he was with some firm of chemists, and I gathered that Sydney Smythe was enjoying the spoils of aristocracy as cashier of his father's bank, but I was not anxious to see Sydney. The others had scattered or married and settled down, and I doubted that they would have much in common with footloose Jamie Abercrombie, who had too much money for his own good and had just inherited another slice that he hadn't earned.

I had dinner and a pipe and then set out along the well-remembered, maple shaded lane of Spring Street, past the old Sutherland place at the corner of Eagle, where a scrawny hedge had replaced the old white picket fence; over the limestone bridge across the Grooterkill, built by one of the Irish stonecutters who had been brought over to work on the Erie Canal; past the Jones house with its neat lawn and big red barn. There was someone on the porch, but I didn't stop. I didn't cotton much to Abigail, and there would be plenty of time in daylight to talk to Mrs. Jones.

Melchizedek Hobbs lived almost at the end of Spring Street, in a huge, rambling clapboard house that hadn't been painted since before Gettysburg. The grass, as usual, was rank on the lawn, but the flower-beds that lined the

flagstone walk were pictures of tender care, and the big new greenhouse in the backyard shone like silver in the moonlight.

There was a light out there, so I went through the side yard and around the house. There was a high wire fence across the yard, with an iron gate, and the gate was padlocked. I rattled it and hooted. The light went out in the greenhouse, and a moment later I saw the gaunt, scarecrow figure of Melchizedek Hobbs stalking toward me.

He knew me at once, in spite of my handsome, flowing moustache and weather-beaten complexion, and after fifteen years. Nor had he changed much himself. He was a bit thinner and he had taken to a pretty obvious toupee. His nose seemed longer and sharper, and a little redder, and his clothes were a little shabbier than I remembered them. He was wearing a butterfly-wing bow-tie instead of the magnificent mauve cravats that I remembered, and it was on crooked.

We went around to the front porch and sat in the summer moonlight, with the mingled perfume of hundreds of flowers wafted up to us from his garden, and the moist, rich smell of the Mohawk in the days before factory wastes and oil tankers turned it into an open sewer. We talked about old times, and about my adventures in far lands, and the exploits of others among his favorite pupils, but I could see that he was uneasy. So, very gradually, I turned the conversation to himself and his flowers. I told him of my experiences in finding some of the plants I had sent him, and he went into raptures over the things he had done with them. And then I asked about the Zulu rose.

It was like throwing a blanket over a coop of clamoring ducklings. I knew he was looking at me through the darkness, his long nose quivering with indecision. I knew that he wanted me to leave, or change the subject, but I knew that he would never ask me to do so. It was cruel, perhaps, but I simply sat and waited.

It seemed a long time before I heard him sigh. "Yes, James. Of course. You have been told something in the village. It was Jim Selford, I presume—he would be the one. Well—you have the right to know."

He got to his feet, and to my amazement began to pull off his coat. He dropped it at his feet and proceeded further to haul his shirt-tails out of his high-waisted trousers. Then, with trembling fingers, he struck a match and held it over his head.

He had on a kind of smock or cassock that came clear down to his bony knees. To the waist it was literally patched with little pockets, and every pocket was stuffed with rich black dirt out of which rose the leaves and stems of seedling plants in various stages of maturity. Some were no more than green buttons and some were well leafed out. Some were flourishing vines, that wound affectionately around his arms and his scrawny neck, and thrust tender tendrils down inside his celluloid collar.

If that was the way he went about, no wonder the town thought he was crazy!

He said nothing. He went down the steps and around through the yard to the greenhouse, and I followed. He unlocked the door and opened it, and I was stifled by a blast of tropical heat and fragrance that sent me winging back to Madagascar and the girl in the hotel.

He stalked down the long aisle of the greenhouse, and I was right at his heels. He lighted lamp after lamp, and as the place filled with light my jaw began to drop, until I must have looked like a candidate for the booby-hatch myself. It was incredible!

The place was full of Zulu roses of every size and description. There were thousands of them—all different—and they filled the greenhouse with a riot of fragrance and rich color that made my head spin. Then I saw something that sent cold fingers diddling along my spine, for as Melchizedek Hobbs walked down the aisle between the banks of plants their gaudy blossoms turned on their stems to follow him, their leaves and stalks stretched out to touch him, and a soft, expectant rustle went up from thousands of straining fibres.

He stopped at a second closed door. "These are the breeding beds and nurseries," he told me. "You are, of course, aware that reproduction in the Zulu rose is bi-sexual and that it does not take place until maturity. There were no male plants among those you sent me, but we have a number of them now."

He opened the door. The greenhouse was L-shaped, and we stepped into a kind of vestibule at the angle. A new perfume flooded into my lungs. I felt my heart pounding, the blood rushing through my veins. I sucked the infernal stuff into my lungs and knew that I was breathing faster, my nostrils dilated, my eyes bright. I remembered a neat pair of ankles I had glimpsed from the cab on Fifth Avenue. I remembered the curve of a dark cheek—the quirk of a pair of soft red lips—the sidelong glance of black eyes. The stuff was an aphrodisiac of the most violent sort, and I saw the color come to Melchizedek Hobbs' pale cheeks and his nose twitching with emotion. He reached up and patted his toupee into place.

He pointed. The plants were growing in pairs, male and female, and their shameless behavior made me gasp. It was outrageous! It was incredible! It was against Nature!

Such abandoned love-making I have never seen in man, beast or bird, let alone a vegetable—and I have seen more than most. The twining stems—the caressing leaves—the squirming, kissing blossoms: I was staring like a silly girl. It was all in the most sensuous of slow-motion, for the things could move as they pleased, or very nearly so. It was like an underwater ballet, completely shameless and completely animal, and I wondered whether any of the town fathers had seen it. If they had, I suspected, Melchizedek Hobbs wouldn't be going about as he was. He'd be in jail, or riding down the turnpike on a rail with a coat of tar and feathers.

The old duffer cleared his throat with a mournful sort of cough. I suspected that he was completely embarrassed. "You see?" he said plaintively. "These creatures are very near the animal in many respects, although they are botanically true plants. They have many traits which I had never thought to find in the vegetable kingdom. You may remember my remarks on that subject, from your school days."

He stared long and gloomily at the rioting blossoms, then cleared his throat nervously. "Eh, yes. These are my young adults, just at the mating age. They are grown in the outer beds, which you have just seen, and brought here when the female begins to mature. The—ah—pollination takes place,

as you see, with much demonstrative display on the part of both sexes. I find it closely akin to the nuptial display of certain pheasants, although there are other aspects—but no more of that. The plants are long-lived, and they will enjoy a—ah—happy wedded life for some weeks, until the young plants begin to bud. Then the male is ignored, his—ah—wooing reflexes degenerate, and he withers away within a night."

He made his way between the beds of oblivious lovers. They were too intent on the business of life to sense that he was there. He opened still another door.

I heard the rustle of leaves as we stepped inside. It was hostile—alarmed—like the buzz of a rattler's tail among dead leaves. He lit the lamp, and I saw that every flower-face in the place was turned toward us. I saw more: their leaves were hugged up like shielding arms, wrapped around their stalks just below the great blooms. There was something alive under those clinging leaves—something small that moved.

Melchizedek Hobbs had taken up a watering can and an artist's palette with little cups of chemicals instead of paint. He went down the aisle, moistening the soil around one plant, stroking another's trembling leaves, feeding a third with lime or potash or some other stuff from the palette. Gradually their leaves unfolded and I saw the little new plants budding from their mothers' stems, just above the highest whorl of leaves. The shape the things took seemed to depend on the kind of soil they were in, but the young plants were all alike, tiny and green and shapeless, much like the embryo of any animal.

Professor Hobbs came back and set down his watering can and palette. His pale eyes were pleading with me to understand. He looked like some medieval sorcerer in his long black robe with its scores of little pockets stuffed with growing plants.

"They are very like animals," he repeated morosely. "The female of the species is quite essential to the normal upbringing of the young. It is not so much a question of nourishment, especially after the young plants have fallen off and taken root, but there is a strong—*rapport*, your French friends would say—between the parent plant and her offspring. Affection, almost. I am convinced that she teaches them the things that they must know to live in the environment in which they find themselves." His eyes were

beginning to gleam. "It is very interesting! Very! I have placed young plants in entirely different soil, fed them entirely different salts, yet so long as they are near their mother they will endeavor to take her form. I have brought stranger-young to a bearing female and placed them among her brood, and they become like her. These—", he touched the tiny plants in his pockets tenderly—"these are orphans which no other plant would adopt. I have had to do so myself."

My head went around like a teetotum. The whole thing was a nightmare! Certainly I had never suspected what would follow my innocent gift of the beautiful flowers which had attracted me so in Madagascar. No wonder the town thought him mad!

We went back through the long greenhouse, and again I saw stems and blossoms twist and sway to greet him. He touched one gorgeous purple bloom and it stiffened under his hand like a cat, but with the slow, painful motion of something which has no right to move.



He touched one gorgeous blossom and it stiffened under his hand like a cat!]

"These are all my children," he said softly. "My first-born." He glanced at me apologetically and his face was flushed. "I must appear odd," he said. "You see, as I have told you, there were no male plants in the bundle which you sent me, and consequently, although it was not difficult to bring them to maturity, pollination of the female flowers was impossible. As soon as I understood a little of their morphology and metabolism I realized that they must be artificially fertilized if the strain was to continue. Lacking the male element, it was necessary for me to devise some mixture of chemicals

which would serve as a substitute. Needless to say, I was successful, and these lovely creatures are the result.

"The methods of insemination which I was forced to employ were drastic in the extreme, I am afraid, but it will never again be necessary to make use of them. We have a fine new generation of young plants growing up and maturing, ready to mate and bring forth their own kind as you have seen. Many of the parent plants, alas, failed to survive. Some of the young died, too, but these you see here I brought up myself, with the aid of one strong plant which did endure my treatment. She is still alive, and these—the children of my science—the young whom I fed through infancy and taught as I once taught you, James—they look to me as to a father. They love me, James. They—and she—and no one else. It has been lonely."

We went back to the house. The cloying perfume of the weird plants still clung to us, and I could see the tendrils of the little "orphans" creeping and writhing over his cassock.

We went inside. It was as I remembered it, fifteen years before—not a picture or stick of furniture had changed. But there was one addition. On the taboret beside his chair, at the left of the great tiled fireplace, was a squat black urn, and in it—the plant.

I realized, of course, that this was the one remaining plant of those I had sent him—the veteran of his experiment—the "she" of whom he spoke. It was showing signs of age. Its waxen leaves were splotched and greyish. Its silky crimson petals, deepening to scarlet at the heart, were faded. Not until he sank down in the old Morris chair and stretched his long legs out toward the hearth did it respond and bend down toward him.

He cradled the great blossom for a moment in his palm, and let his fingers slip lovingly down its slender stem. I saw its withered leaves tremble at his touch, and smelled the faint perfume that rose from it.

"She is growing old, James," he said wistfully. "She is sick and old, and I am all she has. She is very like me, in many ways, and her company has been good for me, but some day soon I must kill her, quickly and painlessly, before disease cripples her any further. It will be the kind thing to do."

I was all wound up inside. They were right in the town—this was mad, abnormal, unhealthy—but he had every reason to be as he was. A man

wholly wrapped up in his science, lonely and misunderstood, suddenly confronted by these exotic, almost animal blossoms: no wonder his curiosity and imagination had been aroused—no wonder solicitude had become something like affection. And in their turn, I realized, these strange plant-animals had learned to look to him for the things which Nature, in this environment, did not provide. They were amazingly quick to adapt: I had known that from the first. So it was that when he fertilized them, taking the place of the missing males, the female flowers accepted him and gave him the weird affection which Nature stored up in them for their normal mates.

That affection, in Nature, assured the species of continued life. It was a blind mechanism, designed by evolution to defy drought and disease and famine. Nature has implanted it very strongly in most animals, but rarely in plants. The female plants looked on him as a mate; the young buds, in their turn, found in him a parent. Oh, it was all very simple to explain in terms of biology and psychology—except to thick-headed, well-meaning village folk of the kind that live in Springville, N.Y. They thought him crazy now, but they would think worse than that if I ever breathed a word of the truth in his defense.

There isn't much more, as it happens. What it was—a hunch—some flash of intuition—maybe the common sense I am supposed to have inherited from my Scots ancestors, and which has made Charles the figure he is on Wall Street—I don't know. I may have remembered the toupee and the bow tie and a word dropped here and there, and put a few numbers together. But next morning early I went down to Melchizedek Hobbs' little flower shop on Main Street to see Abigail Jones.

Abigail's brother had been my best friend in school, and is today, but she and I had never hit it off. She was a good twelve years older than either of us, and she was the perfect figure of the soured, dessicatedly righteous virgin whom we characterize by the tag, "old maid".

The shop showed plainly the care she devoted to it. Everything was immaculate—painfully so—and the potted plants were trim and crisp, the cut flowers fairly sparkling. I wondered where they came from, for there

had been nothing but the Zulu roses in Melchizedek Hobbs' greenhouse, and then I remembered that the Jones family had had fine greenhouses of their own when I was a boy. That was when two and two made four, and I finally made up my mind.

I told her plainly, in so many words, what the trouble was. I took due blame on myself (and I am sure she has never forgiven me) and did my best to point out in a calm, rational, scientific manner that what had happened was the result of purely natural causes operating in a perfectly logical way. Her face never unfroze, her eyes never as much as glinted, and I don't know to this day whether she did what she did because she wanted to or because she thought it was her Christian duty.

As I say, she heard me out without turning a hair. It was only when a sudden flash of inspiration came to me at the very end, as I was halfway out the door, that I thought I saw a bit of a twist on her prim lips. I remembered then that my uncle had had a very fine, large bull, and I told her so.

What happened that night, was in a sense, tragic. The bull got loose, as it had done before. It rooted and rampaged down the length of Spring Street, breaking through the Sutherland's new hedge, plowing up the Pitkins' dahlia beds, scaring a grey mare and spilling out two spooners in a buggy, chasing Constable Nate Williams up a lamp-post, and topping off the evening by raging through Melchizedek Hobbs' greenhouse from end to end. By the time a posse had ramped through after it, and been chased by it, and hosts of small boys and frantic dogs had followed them and fled before them, the species Zulu rose was extinct in the Western Hemisphere.

I say extinct. Melchizedek Hobbs had come out in his crazy smock to drive the beast off, and it treed him. It tore the robe off him and trampled it to ruin. I know, for I was the one who got him down out of the tree when they had cornered the bull.

The old plant was left, and I have always had to give credit to Abigail, much as I sometimes dislike her, because she let him keep it after they were married, up to the point where it began to shed on her rugs. No woman could do more. He killed it then, quietly. And to this day, though Melchizedek Hobbs still potters around the greenhouses and sits in the back of the new store when Abigail will let him, he has never so much as

mentioned the Zulu rose nor his ill-fated attempt to teach young plants the facts of life.

*** END OF THE PROJECT GUTENBERG EBOOK THE FACTS OF
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